

WEEK 5 ASSIGNMENT

Deep Learning & Neural Networks

Task 1 — Deep Learning Fundamentals

Domain	Artificial Intelligence
Batch	6-Week AI Internship
Task	Deep Learning Fundamentals

1. What is Deep Learning?

Deep Learning is a branch of machine learning that uses artificial neural networks with multiple layers to learn patterns and representations directly from data. Unlike traditional approaches that often require manually designed features, deep learning models can automatically learn useful features during training. Deep learning is widely used in image recognition, speech processing, natural language processing, recommendation systems, and many other artificial intelligence applications. The term “deep” refers to the presence of multiple computational layers between the input and output, with each layer transforming information received from the previous layer.

2. Artificial Intelligence vs Machine Learning vs Deep Learning

Artificial Intelligence (AI) is the broad field of creating systems capable of performing tasks that normally require human intelligence. Machine Learning (ML) is a subset of AI in which systems learn patterns from data and use them to make predictions or decisions. Deep Learning (DL) is a subset of ML that uses multi-layer neural networks to automatically learn complex representations. For example, AI includes intelligent assistants and planning systems; ML includes spam detection and price prediction; DL includes image classification and speech recognition.

3. What is an Artificial Neural Network?

An Artificial Neural Network (ANN) is a computational model inspired by biological neural systems. It consists of interconnected processing units called neurons arranged into layers. During training, the network adjusts its weights and biases so that its predictions become closer to the desired outputs. ANNs can learn both simple and complex relationships between input features and target values.

4. What is a Neuron?

A neuron is the basic processing unit of a neural network. It receives input values, multiplies them by corresponding weights, adds a bias, and passes the result through an activation function. A simplified representation is: $\text{Output} = \text{Activation}(w_1x_1 + w_2x_2 + \dots + w_nx_n + b)$. Here, x represents inputs, w represents weights, and b represents the bias.

5. Layers of a Neural Network

The Input Layer receives the input features. For Iris classification, four inputs represent sepal length, sepal width, petal length, and petal width. Hidden Layers perform transformations on the input data and learn useful patterns. A deep network can contain multiple hidden layers. The Output Layer produces the final prediction; for a three-class classification problem it can contain three neurons.

6. Weights and Bias

Weights determine the importance of individual inputs or connections. During training, the neural network updates these values to reduce prediction error. Bias is an additional learnable parameter that shifts the weighted sum and gives a neuron greater flexibility in learning relationships in the data.

7. Activation Functions

Activation functions introduce non-linearity into a neural network, allowing it to learn complex relationships.

Function	Typical Use	Purpose
ReLU	Hidden layers	Introduces non-linearity and is efficient for many neural networks.
Sigmoid	Binary classification	Produces an output between 0 and 1.
Softmax	Multi-class classification	Converts outputs into class probabilities whose sum is 1.

8. Epoch, Loss and Optimizer

Epoch: One complete pass through the training dataset. **Loss:** A numerical measure of how different model predictions are from the actual target values; lower loss generally indicates better predictions. **Optimizer:** An algorithm that updates weights and biases using loss information. Adam is a commonly used optimizer because it adapts the learning rate during training.

9. Basic Working Flow of a Neural Network

Input → Neural Network → Prediction → Loss → Optimization → Improved Model

Input features are provided to the network, which produces a prediction. The prediction is compared with the actual target using a loss function. The optimizer uses the loss to adjust the model parameters. Repeating this process allows the model to improve.

10. Basic Neural Network Architecture

For the Week 5 practical implementation, the ANN uses: **Input (4) → Dense (16, ReLU) → Dense (8, ReLU) → Output (3, Softmax)**. The four inputs are Iris measurements and the three output neurons represent setosa, versicolor, and virginica.

11. Task 1 Conclusion

Deep learning extends machine learning by using neural networks with multiple layers to learn representations from data. Understanding neurons, layers, weights, biases, activation functions, epochs, loss, and optimizers provides the foundation for building and training neural network models. These concepts are applied in the following tasks to create and evaluate an ANN using the Iris dataset.